

BLACK AND SCHOLES (1973) & MERTON (1974)

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1 Partial Differential Equation

Suppose the underlying asset value follows the geometric Brownian motion:

$$dS_t = \alpha S_t dt + \sigma S_t dW_t^{\mathbb{P}}$$

where α is the mean growth rate, σ is the volatility, and $W_t^{\mathbb{P}}$ is the Brownian motion under the physical probability measure. Then suppose we have an option written on that asset, $V(S, t)$. By Itô's Lemma:

$$\begin{aligned} dV_t &= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} dS_t + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} dS_t dS_t \\ &= \left(\frac{\partial V}{\partial t} + \alpha S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \sigma S_t \frac{\partial V}{\partial S} dW_t \end{aligned}$$

Then let's construct a dynamically hedged portfolio of the option with some position in the underlying asset such that the portfolio is riskless, $\Pi_t = V(S, t) - \Delta S_t$, so that

$$\begin{aligned} d\Pi_t &= dV_t + \Delta dS_t + \Delta \delta S_t dt \\ &= \left(\frac{\partial V}{\partial t} + \alpha S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} + \Delta \alpha S_t + \Delta \delta S_t \right) dt + \sigma S_t \left(\frac{\partial V}{\partial S} + \Delta \right) dW_t \end{aligned}$$

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where δ is the return shortfall¹. Clearly, set $\Delta = -\frac{\partial V}{\partial S}$ to make the portfolio risk-free, i.e. to get rid of randomness introduced by the Brownian motion, to have $d\Pi_t = r\Pi_t dt$. The following is then true:

$$\begin{aligned} r\Pi \equiv r \left(V - \frac{\partial V}{\partial S} S_t \right) &= \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} - \frac{\partial V}{\partial S} \delta S_t \\ rV &= \frac{\partial V}{\partial t} + (r - \delta) S_t \frac{\partial V}{\partial S} + \frac{1}{2} \sigma^2 S_t^2 \frac{\partial^2 V}{\partial S^2} \end{aligned} \quad (1)$$

with r being a continuously compounded risk-free rate. Equation (1) is the Black-Scholes-Merton PDE. This PDE also underpins the corresponding Hamilton-Jacobi-Bellman equation if I were to find an option price using the dynamic programming instead of the equivalent contingent claim analysis employed here.

2 European Call Option Price

Black and Scholes (1973, JPE) solve the PDE by transforming it into the heat-transfer equation known from physics subject to boundary conditions. I personally find the straightforward integration to be a more elegant way of getting the price. I first establish the log-normality of the underlying asset at maturity date, then I use it to derive the option pricing formula.

2.1 Establishing Log-Normality

First I suggest the change in the probability measure so that the discounted payoff from holding the asset follows a martingale. First, define

$$W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \frac{\mu - r}{\sigma} t$$

where $\mu = \alpha + \delta$. Note that it makes economic sense: we essentially subtracted the market price of risk in the Black-Scholes economy. Girsanov's theorem states that $W_t^{\mathbb{Q}}$ is the Brownian motion, then

$$dS_t = (r - \delta) S_t dt + \sigma S_t dW_t^{\mathbb{Q}} \quad (2)$$

Then I introduce the solution to the SDE given in the Equation (2). Note that whatever probability

¹So that the total expected return on the underlying is $\mu = \alpha + \delta$ (e.g. capital gain rate plus dividend yield for stocks). By holding the underlying, we receive the “convenience” yield, while by shorting we are required to pay it (otherwise, no rational investor would enter the corresponding long position).

measure we use here, Equation (1) holds, so it is legit. Let me apply Itô's Lemma to $\ln S_t$ to get

$$d(\ln S_t) = \left(r - \delta - \frac{\sigma^2}{2} \right) dt + \sigma dW_t^{\mathbb{Q}}$$

Integrate the SDE over the time interval (t, T) to get

$$\begin{aligned} \ln S_T &= \ln S_t + \left(r - \delta - \frac{\sigma^2}{2} \right) (T - t) + \sigma (W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}}) \\ S_T &= S_t e^{\left(r - \delta - \frac{\sigma^2}{2} \right) (T - t) + \sigma (W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})} \end{aligned}$$

Importantly,

$$\ln S_T \sim \mathcal{N} \left(\ln S_t + \left(r - \delta - \frac{\sigma^2}{2} \right) (T - t), \sigma^2 (T - t) \right)$$

thus

$$S_T \sim \log \mathcal{N} \left(\ln S_t + \left(r - \delta - \frac{\sigma^2}{2} \right) (T - t), \sigma^2 (T - t) \right) \quad (3)$$

so that the underlying asset value at maturity follows a lognormal distribution, Q.E.D.

2.1.1 A Little Note on the Equivalent Martingale Measure $\mathbb{Q}$

To see why the discounted payoff from holding the asset is a martingale under $\mathbb{Q}$, define the risk-free bond with continuous coupon

$$dB_t = rB_t dt$$

and the “dividend” process

$$dD_t = \delta D_t dt$$

by solving which in a similar manner we find

$$B_{t,T} = e^{r(T-t)}$$

$$D_{t,T} = e^{q(T-t)}$$

Then define the numéraire representing the present value of the payoff from holding the asset, $\tilde{S}_t = \frac{D_t S_t}{B_t}$. Itô's Lemma shows (second-order derivative terms turn zero) that

$$\begin{aligned} d\tilde{S}_t &= \frac{\partial \tilde{S}_t}{\partial B_t} dB_t + \frac{\partial \tilde{S}_t}{\partial D_t} dD_t + \frac{\partial \tilde{S}_t}{\partial S_t} dS_t \\ &= -r \frac{D_t S_t}{B_t} dt + \delta \frac{D_t S_t}{B_t} dt + (r - \delta) \frac{D_t S_t}{B_t} dt + \sigma \frac{D_t S_t}{B_t} dW_t^{\mathbb{Q}} \\ &= \sigma \tilde{S}_t dW_t^{\mathbb{Q}} \end{aligned}$$

which yields

$$\tilde{S}_T = \tilde{S}_t e^{-\frac{1}{2}\sigma^2(T-t) + \sigma(W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}})}$$

taking the expectation

$$\mathbb{E}^{\mathbb{Q}}[\tilde{S}_T | \mathcal{F}_t] = \tilde{S}_t e^{-\frac{1}{2}\sigma^2(T-t)} e^{\frac{1}{2}\sigma^2(T-t)} = \tilde{S}_t$$

where $\mathcal{F}_t$ is the time t filtration². I used the fact that $W_T^{\mathbb{Q}} - W_t^{\mathbb{Q}} \sim \mathcal{N}(0, T-t)$ and applied moment generating function for normal variable to evaluate the expectation. $\tilde{S}_t$ is therefore a $\mathbb{Q}$ -martingale, Q.E.D.

2.2 Pricing Formula

European call option price is given by

$$\begin{aligned} C(S, K, t) &= e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ | \mathcal{F}_t] \\ &= e^{-r(T-t)} \int_0^{\infty} (S_T - K)^+ dF(S_T) \\ &= e^{-r(T-t)} \int_K^{\infty} (S_T - K) dF(S_T) \\ &= e^{-r(T-t)} \int_K^{\infty} S_T dF(S_T) - e^{-r(T-t)} K \int_K^{\infty} dF(S_T) \end{aligned} \tag{4}$$

where $F(S_T)$ is the cdf of the S_T , from Equation (3) we already know that S_T follows a lognormal distribution with the log-location $\ln S_t + \left(r - \delta - \frac{\sigma^2}{2}\right)(T-t)$ and the log-scale $\sigma^2(T-t)$.

²Formally, a σ -algebra set such that $\mathcal{F}_t = \sigma\{\tilde{S}_u : 0 \leq u \leq t\}$. Intuitively, think of it as the all the information available up to time t . In finance incorporating such a conditioning would correspond to the weak form of market efficiency.

Starting from the first integral,

$$\begin{aligned}
\int_K^\infty S_T dF(S_T) &= \mathbb{E}^\mathbb{Q}[S_T | S_T > K] \\
&= e^{\ln S_t + \left(r - \delta - \frac{\sigma^2}{2}\right)(T-t) + \frac{\sigma^2(T-t)}{2}} \Phi\left(\frac{-\ln K + \ln S_t + \left(r - \delta - \frac{\sigma^2}{2}\right)(T-t) + \sigma^2(T-t)}{\sigma\sqrt{T-t}}\right) \\
&= S_t e^{(r-\delta)(T-t)} \Phi(d_1)
\end{aligned}$$

where Φ is the standard normal cdf, and d_1 is a familiar notation from the Black and Scholes. In order to evaluate the conditional expectation I simply integrated over the lognormal pdf with domains $(K, +\infty)$ (see, e.g., Hogg and Klugman, 1984).

The second integral is

$$\begin{aligned}
K \int_K^\infty dF(S_T) &= K(1 - F(K)) \\
&= K \left(1 - \Phi\left(\frac{\ln K - \ln S_t - \left(r - \delta - \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}\right)\right) \\
&= K(1 - \Phi(-d_2)) \\
&= K\Phi(d_2)
\end{aligned}$$

where I simply use the expression for the lognormal cdf. Substituting obtained expressions into the Equation (4), I get

$$\begin{aligned}
C(S, K, t) &= S_t e^{-\delta(T-t)} \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2) \\
d_{1,2} &= \frac{\ln \frac{S_t}{K} + \left(r - \delta \pm \frac{\sigma^2}{2}\right)(T-t)}{\sigma\sqrt{T-t}}
\end{aligned} \tag{5}$$

which is a familiar Black-Scholes-Merton formula. Similar step can be undertaken to get the put price (or just use put-call parity).

Alternative way to handle these integrals is to work in terms of the expectation of $\ln S_T$ given that $\ln S_T > \ln K$, it will require the application of Itô's Lemma and then the normal cdf integration will yield the same result as in (5).